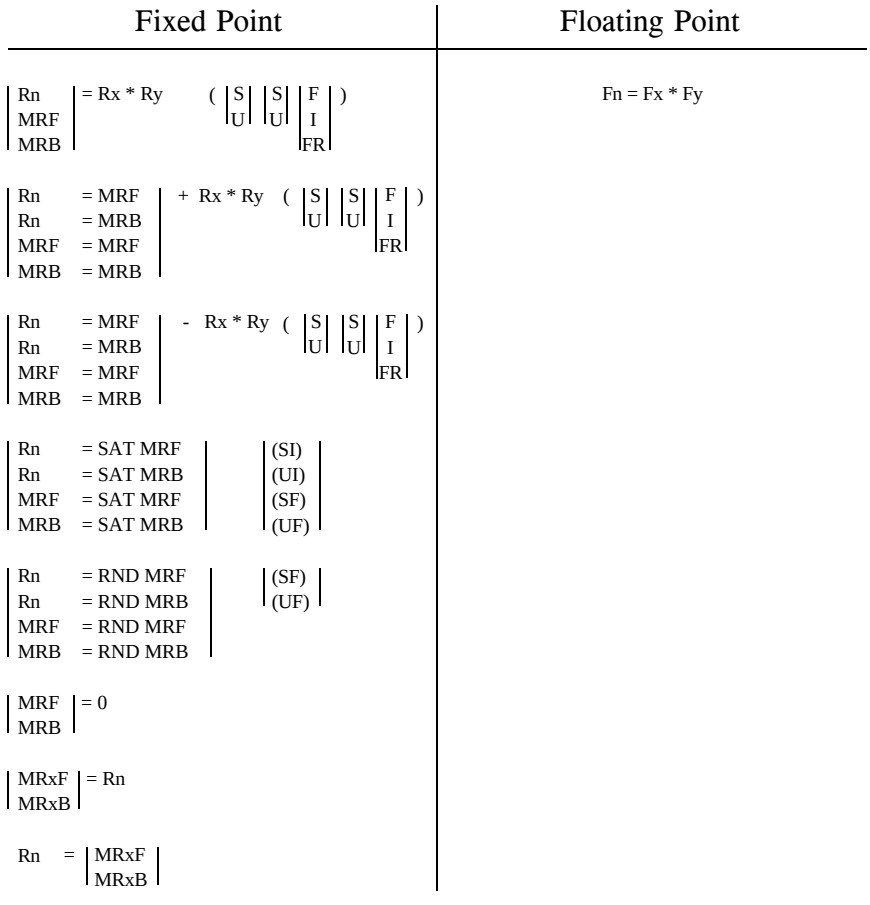

FIGURE 28-7
Fixed versus floating point instructions. These are the multiplication instructions used in the SHARC DSPs. While only a single command is needed for floating point, many options are needed for fixed point. See the text for an explanation of these options.

is, Fn = Fx * Fy, where Fn, Fx, and Fy are any of the 16 data registers. It could not be any simpler. In comparison, look at all the possible commands for fixed point multiplication. These are the many options needed to efficiently handle the problems of round-off, scaling, and format.

In Fig. 28-7, Rn, Rx, and Ry refer to any of the 16 data registers, and MRF and MRB are 80 bit accumulators. The vertical lines indicate *options*. For instance, the top-left entry in this table means that all the following are valid commands: Rn = Rx * Ry, MRF = Rx * Ry, and MRB = Rx * Ry. In other words, the value of any two registers can be multiplied and placed into another register, or into one of the extended precision accumulators. This table also shows that the numbers may be either signed or unsigned (S or U), and may be fractional or integer (F or I). The RND and SAT options are ways of controlling rounding and register overflow.